

VOSTOCHNAYA, Russian Federation

WMO: 246790

Lat: 63.217N

Lon: 139.600E

Elev: 4225

StdP: 12.59

Time Zone: 9.00 (E09)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-49.5	-45.8	-53.1	0.3	-49.1	-49.5	0.3	-45.6	21.2	-25.1	19.5	-26.1	2.2	210	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.6	70.8	54.1	67.2	52.1	64.1	51.0	56.1	65.6	54.3	63.3	52.6	60.9	6.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
52.6	68.8	57.8	50.8	64.3	56.3	48.7	59.5	55.4	25.6	65.7	24.4	63.3	23.3	60.8	63.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.5	18.3	15.7	-52.0	77.8	4.5	3.7	-55.2	80.4	-57.9	82.6	-60.4	84.7	-63.7	87.4		
(5)			-52.0	59.1	4.3	2.2	-55.1	60.7	-57.6	62.0	-60.0	63.3	-63.1	64.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.5	-29.2	-24.6	-10.1	8.1	28.9	46.9	52.2	47.5	31.9	5.1	-17.5	-27.3		
	DBStd	31.19	9.60	9.64	10.43	8.45	8.82	7.09	6.45	6.32	8.57	10.92	10.85	9.81		
	HDD50	14985	2454	2089	1862	1256	655	138	52	120	544	1392	2026	2397		
	HDD65	20254	2919	2509	2327	1706	1120	543	399	542	994	1857	2476	2862		
	CDD50	208	0	0	0	0	0	45	119	43	1	0	0	0		
	CDD65	1	0	0	0	0	0	0	1	0	0	0	0	0		
	CDH74	29	0	0	0	0	0	9	20	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	5.8	4.5	4.9	5.9	6.6	7.0	6.7	5.9	5.7	5.8	6.0	5.6	4.3		
	PrecAvg	11.8	0.1	0.1	0.1	0.4	0.8	2.1	2.6	2.1	1.6	0.6	0.3	0.2		
	PrecMax	16.2	0.4	0.3	0.3	1.0	2.3	4.3	8.2	4.3	3.1	1.3	0.7	0.5		
(15) Precipitation	PrecMin	7.3	0.0	0.0	0.0	0.0	0.4	0.3	0.8	0.7	0.6	0.0	0.0	0.0		
	PrecStd	2.1	0.1	0.1	0.1	0.2	0.4	1.1	1.7	0.9	0.7	0.3	0.2	0.1		
	DB	-0.3	4.2	18.5	34.5	55.8	74.8	76.8	71.3	57.3	31.3	16.6	0.3			
	MCWB	-1.3	3.7	15.6	28.3	42.4	54.5	56.3	54.4	44.6	28.7	15.3	-0.7			
	2% DB	-7.1	-1.9	14.9	29.9	51.0	69.3	72.3	67.1	53.2	27.8	7.7	-4.8			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	-7.5	-2.4	12.5	25.3	39.5	51.9	55.6	52.2	42.7	25.4	6.8	-5.3			
	5% DB	-11.3	-7.1	10.8	25.8	47.6	65.8	68.4	63.8	49.4	24.9	2.7	-8.6			
	MCWB	-11.7	-7.6	8.8	21.7	37.6	49.5	53.6	51.1	40.8	23.1	2.0	-9.2			
	10% DB	-14.5	-11.2	7.1	22.6	43.1	62.0	65.2	60.0	45.6	21.5	-2.0	-12.3			
	MCWB	-14.9	-11.9	5.4	19.2	35.4	48.1	52.4	49.4	38.9	19.7	-2.7	-12.7			
	0.4% WB	-1.7	3.8	16.2	29.2	43.4	55.4	60.4	55.9	46.7	29.3	15.3	-0.8			
	MCDB	-1.1	4.0	18.1	34.0	54.5	70.9	72.3	65.5	53.2	31.1	16.6	0.0			
	2% WB	-7.7	-2.3	12.5	25.2	40.2	52.7	57.3	54.0	43.9	25.6	6.8	-5.3			
(29) Mean Coincident Dry Bulb Temperatures	MCDB	-7.4	-2.0	14.8	29.4	49.2	65.0	66.5	63.0	51.0	27.3	7.5	-5.0			
	5% WB	-11.8	-7.6	8.8	22.0	38.1	51.2	55.8	52.6	41.6	23.1	2.0	-9.1			
	MCDB	-11.6	-7.1	10.8	25.5	46.2	62.2	64.6	60.6	48.4	24.7	2.5	-8.7			
	10% WB	-14.9	-11.7	5.4	19.5	36.2	49.6	54.1	51.2	39.2	19.7	-2.6	-12.7			
	MCDB	-14.5	-11.3	7.2	22.7	43.1	59.6	62.2	58.4	44.8	21.4	-1.9	-12.3			
(35) Mean Daily Temperature Range	MDBR	11.4	15.6	20.8	21.9	16.9	19.3	17.6	18.1	15.4	14.6	12.3	10.5			
	5% DB	13.5	17.5	22.4	23.4	22.8	28.6	25.4	26.2	21.7	13.2	16.7	13.2			
	MCWBR	13.2	17.0	20.0	18.9	13.6	14.0	12.2	13.0	12.9	11.3	16.0	12.9			
	5% WB	13.8	17.4	21.9	21.5	20.1	24.3	21.1	19.6	19.2	12.9	16.5	13.4			
	MCWBR	13.5	16.9	19.5	17.5	12.3	13.0	11.3	11.0	12.0	11.1	15.8	13.0			
(40) Clear-Sky Solar Irradiance	taub	0.159	0.217	0.265	0.322	0.360	0.346	0.391	0.354	0.280	0.229	0.167	0.145			
	taud	1.995	2.169	2.108	1.983	2.039	2.387	2.270	2.360	2.440	2.218	1.959	1.908			
	Ebn at Noon	171	241	265	268	267	277	258	259	264	229	154	98			
	Edh at Noon	9	19	31	44	46	33	36	30	23	17	9	4			
(44) All-Sky Solar Radiation	RadAvg	85	393	994	1668	1818	1685	1558	1197	729	425	145	35			
	RadStd	7	17	34	63	101	143	142	95	74	33	11	3			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page